

Observation of Gravitational Waves from a Binary Black Hole Merger

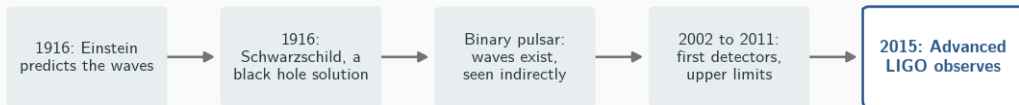
GW150914, seen by both LIGO detectors on September 14, 2015

B. P. Abbott et al.

Conference talk based on Phys. Rev. Lett. 116, 061102 (2016)

LIGO Scientific Collaboration and Virgo Collaboration

Predicted in 1916, never caught directly

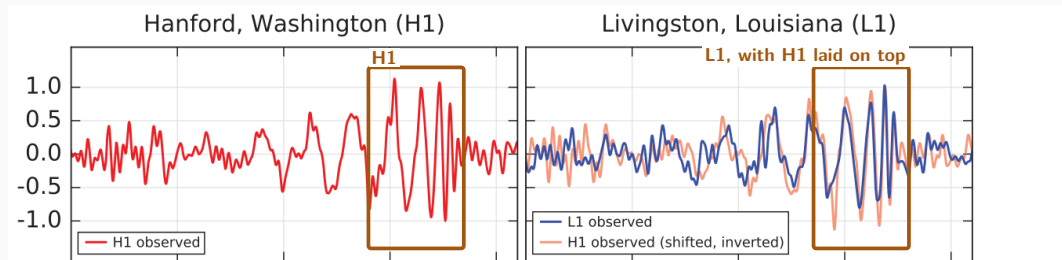


The binary pulsar evidence is indirect: an orbit losing energy, not a wave measured at a detector.

- The waves were known to exist, but **no one had detected one directly, or seen two black holes merge.**

Two detectors, one wave, 6.9 ms apart

September 14, 2015, 09:50:45 UTC: LIGO Hanford (H1) and Livingston (L1)

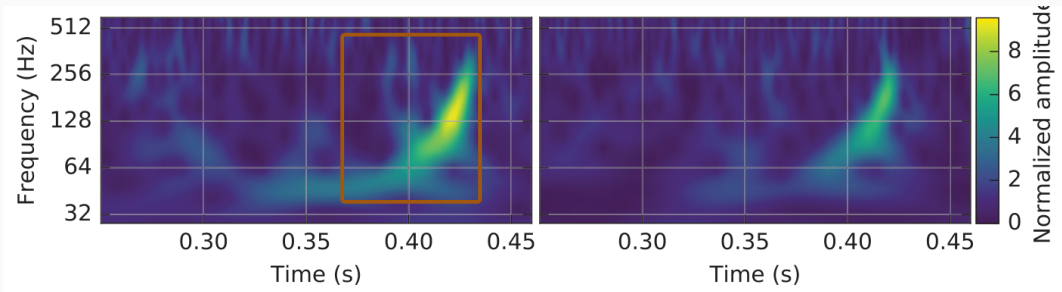


Paper Fig. 1, top row. Strain in units of 10^{-21} , filtered to 35-350 Hz for display.

- The wave reached Livingston first; **shifted by 6.9 ms and inverted**, Hanford's trace lies on Livingston's.

Peak strain 1.0×10^{-21} . Combined matched-filter signal-to-noise ratio: **24**.

Watch the frequency climb



Paper Fig. 1, bottom row: time-frequency map of the strain in H1 (left) and L1 (right).

- In **0.2 s** the signal rises in frequency and amplitude, about **8 cycles from 35 to 150 Hz**, where the amplitude peaks.

What is heavy and small enough to orbit at 75 Hz?

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} = \frac{c^3}{G} \left[\frac{5}{96} \pi^{-8/3} f^{-11/3} \dot{f} \right]^{3/5}$$

1. From the frequency and its rise: chirp mass $\mathcal{M} \simeq 30 M_\odot$.
2. So the total mass is $\gtrsim 70 M_\odot$, and the two Schwarzschild radii add up to $\gtrsim 210$ km.
3. Equal point masses orbiting at 75 Hz would be only $\simeq 350$ km apart.
4. Neutron stars are too light; a black hole with a neutron star would merge at a much lower frequency.

$f, \dot{f}$: observed frequency and its time derivative. Masses in the detector frame. Schwarzschild radius $R_S = 2GM/c^2$. The orbit runs at half the gravitational-wave frequency of 150 Hz.

What is left?

at least 70 solar masses

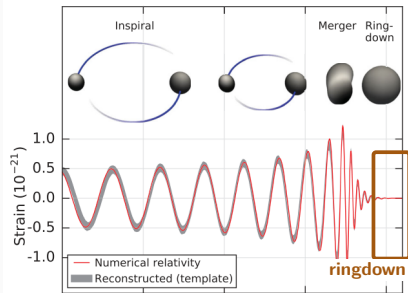


orbiting at 75 Hz, only about 350 km apart



only black holes are that compact

Inspiral, merger, ringdown: the wave general relativity predicts

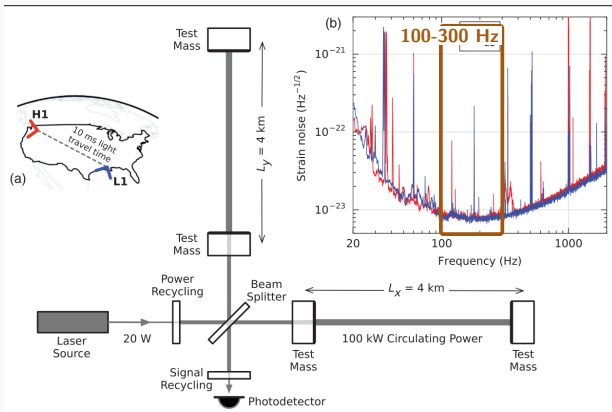


Paper Fig. 2, top: strain in H1 without filtering, with numerical-relativity black hole horizons above.

- After the peak the wave dies away like **a single black hole settling down**, as a Kerr black hole should.

Red: numerical relativity, confirmed to 99.9% by an independent calculation. Grey: reconstruction with black hole templates; in the paper's Fig. 1 it overlaps a model-free wavelet reconstruction by 94%.

The detector is a 4 km ruler for changes in length



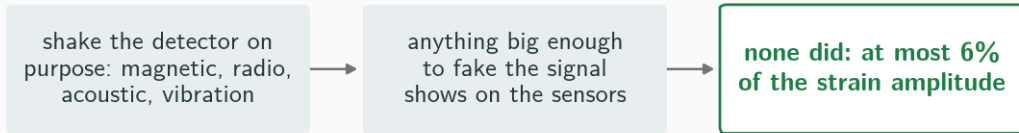
Paper Fig. 3: Advanced LIGO (not to scale); inset (b) is the noise of each detector.

A passing wave changes the arm difference by $\Delta L = hL$, with $L = 4 \text{ km}$.

- Arm cavities multiply the effect on the light phase **300** times.
- **20 W** of laser becomes **700 W** on the beam splitter and **100 kW** in each arm.

Most sensitive band 100-300 Hz: 3 to 5 times better than initial LIGO there, more than ten times better below 60 Hz.

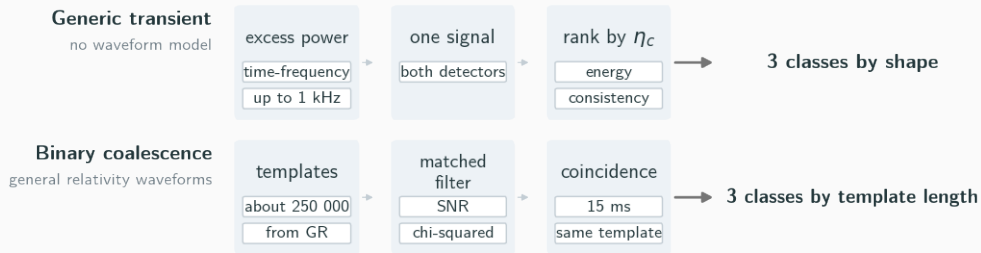
Could the instrument or the ground have made it?



Sensors at each site: seismometers, accelerometers, microphones, magnetometers, radio receivers, weather sensors, power-line monitors, a cosmic-ray detector.

- Both detectors ran steadily for hours around the event, and **no correlated disturbance** between the sites was found.

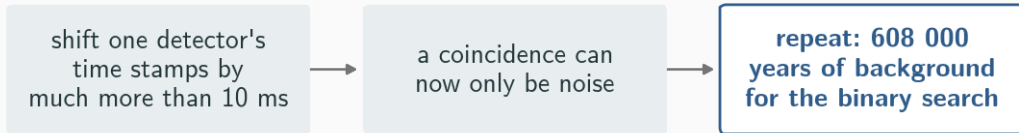
Two searches that assume different things



Independent methods; their noise events are different and uncorrelated.

η_c : coherent signal energy in both detectors, penalised by residual noise, so it measures SNR and agreement at once. $\hat{\rho}_c$: matched-filter SNR, down-weighted when a χ^2 test says the data do not fit the template, added over both detectors. Generic classes: C1 shapes of known noise transients, C3 frequency rising with time, C2 the rest; each event is compared only with background of its own class.

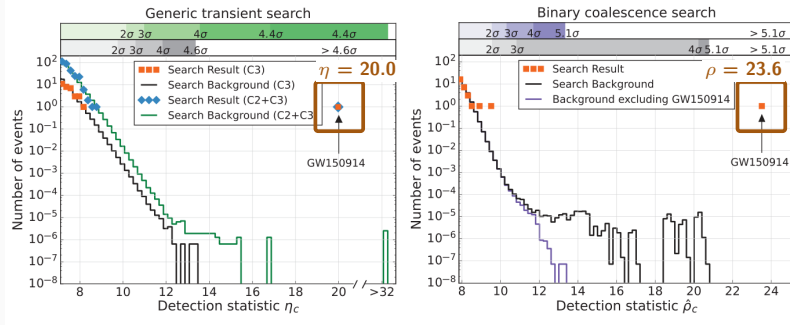
How do you measure the noise when you cannot switch off gravity?



A real signal in one detector can meet shifted noise in the other; that overestimates the background, so the significance is conservative.

- Estimate the background **empirically**: the noise is non-Gaussian and non-stationary, and no shield blocks gravitational waves.

Both searches put the event far out on the tail



Paper Fig. 4: candidates (markers) and mean background (lines); left generic, right binary search.

- In the binary search the event is **louder than any background event**, so only an upper bound on its false alarm rate exists.

Purple: binary background without GW150914; against it the next strongest candidate has a false alarm rate of 1 per 2.3 years.

How often does noise alone make a gravitational-wave signal this strong?

binary coalescence

$> 5.1\sigma$

false alarm rate below 1 in 203 000
years

two independent
searches

generic transient

4.6σ

false alarm rate below 1 in 22 500
years

Confidently detected by both.

Two black holes of about 36 and 29 solar masses became one of 62

Table I	median, 90% interval
Primary black hole mass	36 (+5 / -4) $M_{\odot}$
Secondary black hole mass	29 (+4 / -4) $M_{\odot}$
Final black hole mass	62 (+4 / -4) $M_{\odot}$
Final black hole spin	0.67 (+0.05 / -0.07)
Luminosity distance	410 (+160 / -180) Mpc
Source redshift z	0.09 (+0.03 / -0.04)

Source-frame masses (detector frame = source frame $\times$ (1+z)); errors include waveform-model systematics.

- **3.0 (+0.5 / -0.5) $M_{\odot}c^2$** went out as gravitational waves.

Peak luminosity $3.6 (+0.5 / -0.4) \times 10^{56}$ erg/s, or $200 M_{\odot}c^2$ per second.

Primary spin below 0.7 (90% credible), so not maximally spinning; the secondary's spin is only weakly constrained.

Does it follow general relativity?

Test	What it compares	Result
Final black hole	mass and spin from the early stage vs. the late stage	consistent
Inspiral phase	post-Newtonian coefficients allowed to deviate	no evidence of violation
Graviton	a modified dispersion relation	$\lambda_g > 10^{13}$ km, so $m_g < 1.2 \times 10^{-22}$ eV/ c^2

The graviton bound beats Solar System and binary pulsar bounds, but not those from galaxy clusters and weak lensing.

All three tests are **consistent with general relativity** in the strong-field regime.

What the black holes tell us

- Stellar-mass black holes **heavier than about 25 solar masses exist**, and binary black holes **form in nature and merge within a Hubble time**.

Merger rate: **2 to 400 per Gpc³ per year**, depending on the mass model; only the lowest predicted rates are excluded.

Forming such heavy black holes from stars requires weak stellar winds, possible below about half the solar metallicity.

What two detectors cannot tell us yet

Limits

- With only two detectors the source is localized to about **600 deg²**.
- The secondary black hole's spin is only **weakly constrained**.
- These are **16 days** of data; the full first run is reported later.

Next

- Design sensitivity: binaries like this one at **3 times** the SNR.
- Advanced Virgo, KAGRA and a possible LIGO detector in India improve position and parameters.

The first direct detection of gravitational waves, and the first observation of a binary black hole merger.

Backup: every search puts GW150914 above 4.4 sigma

Search	Statistic	Background	False alarm rate	Probability	Significance
Binary coalescence	$\hat{\rho}_c = 23.6$	608 000 yr	< 1 in 203 000 yr	$< 2 \times 10^{-7}$	$> 5.1\sigma$
Generic, class C3	$\eta_c = 20.0$	67 400 yr	< 1 in 22 500 yr	$< 2 \times 10^{-6}$	4.6σ
Generic, C2+C3	$\eta_c = 20.0$	same events	1 in 8 400 yr	5×10^{-6}	4.4σ
Second event (binary)	-	without GW150914	1 per 2.3 yr	0.02	-

C3 includes a trials factor of 3. C2+C3 has four background events at 32.1 or more. A dash: the paper gives no value for that cell.

Backup: the detector numbers behind the sensitivity

Quantity	Value
Arm length	4 km
Test masses	40 kg fused silica
Seismic isolation above 10 Hz	more than 10 orders of magnitude
Beam tube pressure	below 1 μPa
Calibration (1 sigma)	under 10% amplitude, 10 degrees phase
Timing	GPS to better than 10 μs